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PS – 389

II Semester M.C.A. Examination, November/December 2022
(CBCS) (2020-21 and Onwards)
COMPUTER SCIENCE
2MCA4 : Software Engineering

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Section – **A** and **any four** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Compare and contrast the traditional and agile methods of software engineering.
2. List 3 important characteristics of agile software teams and discuss the need to assign roles for each team member in an agile project. What are the various role schemes in the Agile ?
3. Discuss the practices in Agile that support customer collaboration and user-centric development.
4. Explain the terms :
   a) Burn-up and burn-down charts
   b) Sustainable pace
   c) Team velocity with suitable examples.
5. Discuss ways in which software quality is expressed in software development environments.
6. Define constructivism. How does this approach support learning in the agile environment ?
7. What is Abstraction ? How is it expressed ? Is it important in software engineering ? Explain.
8. How can ethics be expressed in software engineering processes ? Illustrate your ideas with specific scenarios.

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SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. State the Agile Manifesto. Explain the basic practices in Agile Software Development and analyze them from the HOT perspectives.

10. a) Give the use case diagram and sequence diagram for a Bank ATM.
    b) Discuss dilemmas and rewards in team work.

11. a) Discuss the activities that take place in a Business Day. How is it different from a stand-up meeting ?
    b) What are the techniques used to ensure tightness of software projects ?

12. a) What are measures in Agile ? Discuss the various measures that suit large scale agile projects to verify that the goals of the project are met.
    b) Discuss the pros and cons of Test Driven Development. How does TDD help to solve inherent problems in testing ?

13. a) Choose any three agile practices and explain how they guide practitioners to move between abstraction levels during the development process of a software project.
    b) Suggest any 2 ideas from game theory that can be utilized for the understanding of software development in general and agile software development in particular.

14. Discuss the following :
    a) Describe your cell phone on any 2 levels of abstraction.
    b) Pair programming.
    c) Refactoring.
    d) User Centric Design.

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II Semester M.C.A. Examination, November/December 2023
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA4 : Software Engineering

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any 5 full** questions from Part – **A** and **any 4 full** questions from Part – **B**.*

PART – A

Answer **any 5 full** questions. **(5×6=30)**

1. State and elucidate Agile Manifesto in detail. **6**
2. Give a description of role scheme in Agile teams. **6**
3. Write a short note on pair programming, refactoring, refinement. **6**
4. Depict the concept of sustainable pace and also write its importance. **6**
5. How to achieve tightness of agile methods ? Explain in detail. **6**
6. Designate the role of short releases and iterations in learning process. **6**
7. Categorize delivery and cyclicality in learning environments. **6**
8. Illustrate advantages of global software development in detail. **6**

PART – B

Answer **any 4** questions. **(4×10=40)**

9. With the help of example illustrate use case, class, interaction and activity diagram. **10**
10. Give details of Agile software development in learning environments. **10**
11. List and narrate the time related problems of software projects. **10**
12. How does test driven development overcome some of the problems inherent in testing ? **10**
13. Characterize role scheme in Agile software team and also designate role activities. **10**
14. Write a short note on :
    i) Reflective practitioner perspective. **5**
    ii) Retrospective. **5**

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II Semester M.C.A. Examination, December 2024
(CBCS Scheme)
(2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA4 : Software Engineering

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **all** the Sections.*
*2) Answer **any five** questions from Section – **A** and **any four** questions from Section – **B**.*

SECTION – A

I. Answer **any five** of the following. **Each** carries **6** marks. **(5×6=30)**

1) Differentiate between traditional and Agile software development.
2) Explain the role of scheme in Agile software team and also explain different role and its activity.
3) What measures can be used for time management in agile projects ?
4) How would you suggest maintaining software quality for a specific software project ?
5) Explain short releases and iterations in learning process.
6) Explain organizational culture in agile distributed team.
7) Discuss delivery and cyclicality in learning environments.
8) Describe the different leadership styles that empower the people involved in the project.

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SECTION – B

II. Answer **any four** of the following. **Each** carries **10** marks. **(4×10=40)**

9) a) Give a detailed explanation of the Agile Manifesto. **5**
b) Write a note on test-driven development. **5**

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10) a) Draw a use case diagram for college management system. **5**

b) Draw a sequence diagram for ATM. **5**

11) a) Discuss the activities that take place in a business day with the practice of short iterations and releases. **5**

b) What qualities are implied for high-quality process ? **5**

12) What is reflective practitioner perspective and retrospective ?

13) Define abstraction. Explain the abstraction levels in agile software development.

14) Explain in detail conceptual framework for change.

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II Semester M.C.A. Examination, November/December 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER APPLICATION
2MCA4 : Software Engineering

Time : 3 Hours

Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five**.*
*2) Part – **B** : Answer **any four**.*

PART – A

Answer **any five** of the following questions. **(5×6=30)**

1. With an example, explain the state chart.
2. Explain the characteristics of Agile Software Development.
3. Discuss the time related problems of software projects.
4. Write a note on Refactoring and Test Driven Development.
5. Explain the role of TDD in overcoming the problems inherent in Testing.
6. Explain the application of Agile practices in Retrospective Sessions.
7. Discuss the application of Agile principles in Non-software Projects.
8. Explain the benefits of short iterations linked to learning process.

PART – B

Answer **any four** of the following questions. **(4×10=40)**

9. Discuss the waterfall and agile software development approach with the merits and demerits.
10. Explain the student management system with the help of use-case diagram, class diagram and interaction diagram.

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11. Explain the Game Theory perspective in software development.

12. Explain Agile Manifesto in detail.

13. a) Explain the role scheme in Agile practices. 5

    b) Discuss the importance of tracker help in improving the quality of the overall software developed. 5

14. a) Discuss the importance of measures in agile software development. 5

    b) With an example, describe the measures taken to ensure that the software delivery happens on time with the right quality. 5

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